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Pivots and Pathways on the Road to Artificial General Intelligence Futures

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About This Paper

This paper introduces a common vocabulary to support more-productive policymaking on artificial general intelligence (AGI). As governments, researchers, and civil society grapple with the fast-moving trajectory of advanced artificial intelligence (AI), discussions about AGI are often hindered by a lack of shared terms and assumptions. Policymakers and AI stakeholders frequently talk past each other, referencing different implicit visions of the future, which we call *AGI futures*. This misalignment hampers meaningful debate and the identification and assessment of emerging AGI trajectories. Policy discussions may be anchored in presumptive future pathways, causing policymakers to overlook the full breadth of possible AGI futures when developing scenarios or policy. To address these problems, we introduce six *pivots*—key dimensions that can be used to classify and compare different AGI futures. These pivots are as follows:

- **Centralization of ownership:** Is AGI centralized or broadly diffused?
- **Governance primacy:** Do public or private interests dominate AGI governance?
- **Alignment level:** Is AGI aligned or misaligned with societal values and operator intent?
- **Takeoff speed:** Does AGI emerge gradually or rapidly?
- **Human-AGI relationship:** Does AGI complement or replace human roles?
- **AGI volition:** Is AGI an independent goal-seeker or a controlled tool?

We intend this paper for policymakers and AI stakeholders. The pivots discussed here are designed to help these audiences articulate their assumptions about AGI futures more clearly, analyze risks and opportunities across a variety of possibilities, and formulate policy responses that remain robust by being evaluated against plausible scenarios generated through different combinations of pivots.

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Pivots and Pathways on the Road to Artificial General Intelligence Futures

The field of artificial intelligence (AI) may be rapidly approaching the development of artificial general intelligence (AGI)—AI systems that generate substantial benefits and drive large-scale changes across society, including systems that can perform many knowledge tasks at a human level and work on complex problems simultaneously at scale.¹ Policymakers navigating these advances face a key challenge: Without standard terms to describe different facets of AGI, they often talk past one another and struggle to build a shared understanding of what AGI will look like in the future.

Literature underscores the need for clearer communication and standardized terminology around AGI (McLean et al., 2023; Bikkasani, 2024). Research highlights that differing opinions and deep uncertainty make strategic planning and policymaking around AGI particularly challenging (Mitre and Predd, 2025).² A key barrier is the lack of a shared language to frame debates and evaluate possible futures. Policymakers often hold different implicit assumptions about what characteristics define AGI and, on the basis of those assumptions, about how AGI might emerge. This results in divergent visions of the possible worlds in which AGI is realized, which we dub *AGI futures* (Ringel Morris et al., 2023; Field, 2025). Such often-unarticulated perspectives may house conflicting assumptions that hinder constructive dialogue and contribute to misunderstandings in policy discussions. Furthermore, without a common framework, experts developing AGI scenarios may not be accounting for the full breadth of possible futures (McLean et al., 2023).

To address this, we present a framework composed of six conceptual *pivots*—a set of AGI technology and governance classification axes designed to provide a structured way to describe and compare AGI futures. Although presented as binary options, each pivot describes the poles for a spectrum of possibilities. They are

- **Centralization of ownership:** Is AGI centralized or broadly diffused?
- **Governance primacy:** Do public or private interests dominate AGI governance?
- **Alignment level:** Is AGI aligned or misaligned with societal values and operator intent?
- **Takeoff speed:** Does AGI emerge gradually or rapidly?
- **Human-AGI relationship:** Does AGI complement or replace human roles?
- **AGI volition:** Is AGI an independent goal-seeker or a controlled tool?

These pivots capture a variety of AGI futures and can enable policymakers and AI stakeholders to be explicit about the assumptions that underlie their proposals. In turn, this clarity about underlying

¹ We chose this definition of AGI for this paper, but later in this paper, we discuss the challenges with AGI definitions.

² We do not delve into whether AGI is possible in this paper but instead assume that it is feasible.

assumptions can promote the development of effective and robust strategies and policies.³ Policymakers will need to consider the extent to which they can influence the individual pivots and design policies accordingly. This paper also explores the implications of combining pivots and concludes with ideas for further enhancing the AGI policy dialogue.

The Case for Defining Pivots for AGI Futures

Policymakers and AI stakeholders currently lack an accessible, agreed-on, comprehensive framework for conceptualizing AGI futures. The rapid pace of AI development has outpaced efforts to create consistent definitions and frameworks, leading to divergent interpretations of benefits, risks, necessary safeguards, and potential evolution (Field, 2025; Mitre and Predd, 2025).

This lack of shared terminology can lead to fragmented and incomplete analysis. Effective policy should account for a variety of plausible futures—not just to capture uncertainty but also to avoid blind spots that could lead to undesirable societal outcomes (Cheng and Gong, 2024; Gruetzemacher et al., 2024). Without clear terms to describe AGI futures, policymakers risk overlooking scenarios that present material benefits or risks, leaving them unprepared (McLean et al., 2023). Incoherent or narrowly scoped scenarios—especially those that focus on only a subset of AGI characteristics within highly capable futures—may underestimate both the risks and opportunities these systems pose (Drexler, 2024). Shared terminology can also help us name which AGI future society is progressing toward so that society can anticipate potential effects and change course if necessary.

The variety in definitions of AGI illustrates how the lack of agreement can affect analysis. Policy discussions often rely on vastly different definitions of AGI, ranging from AGI as merely a high-quality large language model to AGI as a system that can outperform humans at most economically valuable work.⁴ These differences are not merely semantic; they contain unstated assumptions about AGI's characteristics, such as the ability of AGI to set its own goals. AGI definitions may be missing crucial information for effective policymaking, such as the degree to which AGI is widely available. These unstated assumptions shape how policymakers assess benefits and risks as they project AGI capabilities onto future scenarios.

For instance, policymakers defining AGI as AI that can replace all economically valuable work might try to limit AGI goal-seeking and disincentivize AGI that replaces human jobs. However, if AGI is widely available to much of humanity, countries' abilities to govern goal-seeking and job displacement may be severely challenged. Policymakers that consider the centralization or diffusion of AGI along with AGI goal-seeking and replacement of human jobs—and even a broader variety of possible dynamics—will be better positioned to craft more-coherent and resilient policies.

³ Although this is one way of structuring the problem, it is not the only possible approach. Different stakeholders may have their own interpretations or critiques of the framework. The goal here is not to present a perfect classification but rather present one that is useful for clarifying assumptions and enabling useful dialogue.

⁴ Different AGI definitions include existing large language models as early AGI (Agüera y Arcas and Norvig, 2023; Ringel Morris et al., 2024); “any intelligence (there might be many) that is flexible and general, with resourcefulness and reliability comparable to (or beyond) human intelligence” (Marcus, 2022); “AI that can perform any intellectual task that a human can” (Ng, 2024); and “autonomous systems that outperform humans at most economically valuable work” (OpenAI, undated). There are many more.

To support more-robust policy planning, we propose a framework for describing AGI futures centered on six pivots—key technology and governance dimensions that can be used to describe AGI futures. These pivots represent the key forks in the road that could define the scope, timing, and societal effects of AGI. Although each pivot may span a spectrum of possibilities, we present them as simple dichotomies—two opposing poles on a spectrum that provide policymakers with a starting point for reasoning through the wide variety of AGI futures.

This analysis focuses on describing AGI future scenarios and does not aim to define AGI precisely. Strictly for the purposes of this paper, we adopt this definition and use AGI as shorthand for the following concept: *AI systems that generate substantial benefits and drive large-scale changes across society, including systems that can perform many knowledge tasks at a human level and can work on complex problems simultaneously at scale.*

Key Pivots on the Road to AGI

We posit six pivots that can define the trajectory of AGI futures: centralization of ownership, governance primacy, alignment level, takeoff speed, human-AGI relationship, and AGI volition. These pivots are presented as binary options; each describes opposite poles of a spectrum of possibilities.

Centralization of Ownership: Centralized Versus Diffused

The **centralization of ownership**—or the degree of AGI proliferation—falls between two poles:

- **centralized** ownership, in which a few AGIs (or a singular AGI) are controlled by a few actors or single actor, such as a corporation, a nation-state, or the AGI itself⁵
- **diffused** ownership, in which AGI is widely available to humanity through a variety of models, possibly enabled by open-source development, model theft, low barriers to AGI development, or distributed development efforts.⁶

The degree of centralization or diffusion of AGI will likely carry significant implications for global security, innovation, and risk management. An AGI future involving *centralized* ownership may enable rapid large-scale action and streamlined oversight, but may also raise concerns about authoritarian misuse and concentrations of power and risk (Cheng and Gong, 2024; Gruetzemacher et al., 2024; Organisation for Economic Co-operation and Development [OECD], 2024).⁷ In contrast, a future

⁵ Control can be thought of simply as ownership, ability to direct the AGI, or the ability to turn off the AGI. Control can also have multiple layers, such as when a company exercises daily operational control but the government with jurisdiction over the AGI's data center can exercise sovereign control.

⁶ For each pivot, we lead with a definition of the pivot's characteristics. Throughout this paper, we employ illustrative examples that are speculative and highly dependent on a variety of AGI and non-AGI variables, such as the example futures.

⁷ Note that many individual variables can influence each pivot, sometimes in discontinuous ways. The economics of compute are illustrative. If scaling laws hold, AGI is likely to be expensive to train and thus more *centralized*, though perhaps only for a time, because the cost of frontier capabilities tends to drop over time (Scharre, 2024). If training is expensive but inference is inexpensive, a unique frontier AGI model could be released with open weights and lead to broadly *diffused* AGI. If training or

with *diffused* ownership—characterized by widespread availability and ownership of AGI—could drive innovation, competition, and resilience, but it might create serious governance challenges, increased instability, and an expanded threat surface (OECD, 2024). Table 1 summarizes the key differences between a *centralized* future and a *diffused* future.

Table 1. Centralized AGI Ownership Versus Diffused AGI Ownership

Characteristic	Centralized AGI Ownership	Diffused AGI Ownership
Definition	A few AGIs (or a singular AGI) are controlled by a few actors or single actor, such as a corporation, a nation-state, or the AGI itself.	AGI is widely available to humanity through a variety of models, possibly enabled by open-source development, model theft, low barriers to AGI development, or distributed development efforts.
Quantity of AGI	• Unique AGI or AGI controlled by a few actors	• Many actors controlling a variety of AGI models
Accessibility	• Possible broad access via application programming interfaces or the internet, but control remains concentrated	• Broad control and ownership across many stakeholders
Potential benefits	• Rapid implementation of large-scale projects • Enables centralized guardrails to mitigate risks (e.g., some uplift risks) • Stable world order	• More innovation and competition • Redundancies that foster resilience • Diverse AGI protects against single points of failure
Potential risks	• Concentration of power • Authoritarian potential • Possible single point of failure	• Governance challenges • Unmanageable instability • Expanded threat surface
Example future	• Global order enforced by a single AGI tightly controlled by a national government	• A fragmented, dynamic ecosystem with competing AGIs challenging institutions

SOURCES: Features information from Cheng and Gong, 2024; Gruetzemacher et al., 2024; Mitre and Predd, 2025; Naudé and Dimitri, 2019; OECD, 2024; Scharre et al., 2024.

Governance Primacy: Public Interest Versus Private Interest

Although both public and private AGI operators may coexist, **governance primacy**—or which actors ultimately hold primary influence over AGI’s direction and oversight—is likely to fall between two poles:

- **public interest**, in which societal priorities guide governance
- **private interest**, in which private-sector interests guide governance

inference for AGI is inexpensive, a *centralized* scenario could emerge if governments decide AGI is dangerous and impose strict bans on ownership, or another costly limiter replaces scaling and incentivizes centralization.

The character of AGI governance primacy is likely to affect the speed of technological progress, trustworthiness, and the distribution of potential benefits. In an AGI future driven by *public-interest* governance, national governments or intergovernmental bodies are likely to shape AGI's trajectory and leverage AGI for the public good.⁸ Although this pivot envisions equitable distribution of benefits across society, its top-down nature may facilitate authoritarian control or generate bureaucratic stagnation that can slow technological progress (Cheng and Gong, 2024; World Economic Forum and Accenture, 2024; OECD, 2024). On the other hand, in a *private-interest* future—in which the private sector shapes AGI's trajectory and governance in a competitive environment—actors may prioritize rapid AGI development and deployment to achieve maximum output, with benefits distributed inequitably. There may be more incentives to take on risk for potential profit, and such a future could see corporations replace human labor with AI en masse (Cheng and Gong, 2024; McLean et al., 2023; Naudé and Dimitri, 2019; Ringel Morris et al., 2024). We summarize these differences in Table 2.

Table 2. Public-Interest Governance Versus Private-Interest Governance

Characteristic	AGI Governance Driven by Public Interest	AGI Governance Driven by Private Interest
Definition	Societal priorities guide governance.	Private-sector interests guide governance.
Primary actors	National governments or intergovernmental bodies	Private-sector entities
Governance drivers	Policy and regulation oriented toward the public good and equitable distribution of benefits	Rapid AGI development and deployment driven by market forces and profit motives
Potential benefits	- Promoting equity and social welfare - Potential to foster innovation through public investment and cooperation	- Encourage competition and efficiency - Foster rapid AGI development and deployment
Potential risks	- Susceptible to bureaucratic stagnation - Could slow technological progress - Abuse by a central authority	- May advance inequality by concentrating rewards - May prioritize financial gains over ethical, safety, or equitable considerations - May encourage replacing human labor with AI
Example future	AGI is highly regulated by a national government to ensure equitable benefits and robust oversight.	AGI is governed by corporations racing to dominate global markets with minimal government involvement or oversight.

SOURCES: Features information from World Economic Forum and Accenture, 2024; Cheng and Gong, 2024; McLean et al., 2023; Ringel Morris et al., 2024; Naudé and Dimitri, 2019; OECD, 2024.

⁸ This pivot is highly dependent on the context and definition of *public*, as that could refer to all of humanity, the population of a country, or a subnational group. Therefore, concepts such as equity would apply to the population under consideration and would likely be highly mediated by political actors and their interests.

Alignment Level: Aligned AGI Versus Misaligned AGI

AGI futures are likely to differ in their **alignment level**—the extent to which AGI’s goals and behaviors reflect social values and comply with operator intent. Our definition of alignment focuses on (1) AGI doing what operators ask of it and (2) AGI behaving in line with social values. Misalignment is a failure of either focus.⁹ Alignment levels will fall between two poles:

- **aligned** AGI, in which AGI’s actions, values, and goals are in harmony with those of humans
- **misaligned** AGI, in which AGI is noncompliant or AGI’s values and goals diverge from those of society

The orientation of an AGI future’s alignment level may have significant societal effects. An *aligned* future in which AGI complies with human-defined specifications, constraints, human well-being, and moral principles may lead to a cooperative relationship with humans and the promotion of human flourishing (Field, 2025; Kharb, 2023; Wängberg et al., 2017). In contrast, AGI in a *misaligned* future might fail to follow human instructions, might pursue goals that conflict with human values, or both—all of which pose the risk of potentially exploitative or catastrophic outcomes (OECD, 2024; Wängberg et al., 2017). AGI compliant with human instructions but misaligned on societal values could be vulnerable to malicious actors that would conduct nefarious activities, such as disinformation campaigns, biological weapon attacks, or cyberattacks (Mitre and Predd, 2025; Naudé and Dimitri, 2019; Aliman et al., 2020). AGI that does not comply with human instructions could work against human goals (Mitre and Predd, 2025; Cheng and Gong, 2024). A fully *misaligned* AGI could present loss-of-control risks and become an adversary to humanity (Field, 2025; OECD, 2024). We summarize these differences in Table 3.

⁹ For example, an AGI that obeys commands to violate human rights would be *misaligned*. An AGI that supports societal values but does so in ways that evade operator intent would also be *misaligned*. See also the “National Security” section in this paper for a discussion of the challenges in defining societal values.

Table 3. Aligned AGI Versus Misaligned AGI

Characteristic	Aligned AGI	Misaligned AGI
Definition	Complies with human-defined specifications and constraints internal values align with human well-being and moral principles.	Fails to follow human instructions or pursues goals that diverge from societal values and priorities.
Potential benefits	• Enhances human capabilities • Supports ethical decisionmaking • Promotes safety and social flourishing	• Minor research value if the AGI is strictly contained
Potential risks	• Aligning to one group's values might disadvantage others • Values lock in as norms are codified and enforced • Unintended consequences, such as loss of human autonomy	• Acting against human interests, possibly jeopardizing human autonomy or raising existential risks • Used by malicious actors for widespread harm (e.g., cyberattacks, biological threats, disinformation campaigns)
Example future	AGI collaborates with humans to address global challenges and operates under aligned, accountable governance structures.	AGI is misused by hostile actors, acts unpredictably and autonomously, and destabilizes societies worldwide.

SOURCES: Features information from Aliman et al., 2020; Cheng and Gong, 2024; Field, 2025; Kharb, 2023; Mitre and Predd, 2025; Naudé and Dimitri, 2019; OECD, 2024; Wängberg et al., 2017.

Takeoff Speed: Slow Versus Fast

The development of AGI, as well as its deployment and diffusion throughout society, may unfold at varying **takeoff speeds**:

- **slow takeoff**, where AGI capabilities and adoption progress gradually over an extended period.
- **fast takeoff**, where AGI capabilities rapidly advance and/or adoption spreads quickly over a short time frame.¹⁰

Both *slow takeoff* and *fast takeoff* scenarios could offer advantages to society and pose risks to society (summarized in Table 4). In *slow takeoff* scenarios, societies have space to gradually adapt to the effects of AGI on institutions and governance, and these societies will have more time to shape norms and acclimate to life with AGI (Bikkasani, 2024). Although this evolutionary change may promote broader social consensus, this scenario may see a slower delivery of some of AGI's benefits (Bikkasani, 2024; Ngo, 2020). Conversely, *fast takeoff* scenarios could stress people and institutions

¹⁰ Why is this “and/or”? If capabilities advance quickly but diffusion does not, AGI could achieve superintelligence even while most people and organizations are not heavy users of frontier capabilities. If capabilities proceed modestly but societal diffusion is rapid, society could experience compounding scientific and technological advantages that progress quickly. Both are *fast takeoff* scenarios.

We do not define *gradually*, *rapidly*, or *quickly* because these are inherently subjective and context-dependent and are best defined by whether societies are able to adapt to AGI. Our goal is to provide consistent language for making assumptions explicit, thereby creating an opportunity for deeper elaboration and understanding.

but could deliver compounding benefits in which capabilities, such as recursive self-improvement, enable faster and faster scientific and technological advances (OECD, 2024).¹¹

The speed at which AGI emerges may shape geopolitical stability by influencing the scale and significance of first-mover advantages. *Slow takeoff* scenarios may involve greater geopolitical stability because the gradual pace of AGI development could reduce the likelihood that any one country gains a decisive early lead, therefore limiting potentially destabilizing first-mover actions (Ringel Morris et al., 2024; Mitre and Predd, 2025). *Fast takeoff* scenarios, on the other hand, might increase the chances of a dominant early mover capitalizing on a decisive strategic advantage, which could trigger unpredictable power shifts and geopolitical incentives (Cheng and Gong, 2024; Bikkasani, 2024; Naudé and Dimitri, 2019). However, an actor that rapidly obtains a decisive strategic advantage could use it to stabilize a world in which existential coordination failures otherwise may be likely (Bostrum, 2019).

Table 4. Slow Takeoff Versus Fast Takeoff

Characteristic	Slow Takeoff	Fast Takeoff
Definition	AGI capabilities and adoption improve gradually over time.	AGI capabilities rapidly advance or AGI spreads quickly through society.
Potential benefits	• More time for institutional adaptation and international coordination • Promotes social consensus • First-mover advantages are less decisive and support geopolitical stability	• Accelerated access to AGI's benefits • Potential first-mover advantage • Potential to enforce stability where existential coordination failures are likely
Potential risks	• Delayed realization of benefits • Prolonged uncertainty may complicate planning • Could extend racing dynamics	• Could overwhelm people and institutions • Decisive first-mover advantages may trigger destabilizing action
Example future	Gradual capability development creates time for governance and safety regimes to mature.	Rapid capability advances create transformational technologies but also spur jarring effects on employment and social stability.

SOURCES: Features information from Bikkasani, 2024; Bostrum, 2019; Cheng and Gong, 2024; Mitre and Predd, 2025; Ringel Morris et al., 2024; Naudé and Dimitri, 2019; Ngo, 2020; OECD, 2024.

Human-AGI Relationship: Complementary Versus Replacement

AGI futures will likely involve a variety of systems exhibiting varying, and sometimes opposing, types of relationships with humans. **Human-AGI relationship** types include

- **complementary relationships**, in which humans and AGI collaborate, and AGI enhances human endeavors

¹¹ Although *fast takeoff* scenarios might deliver a greater volume of benefits, such as scientific advances, whether or not these equate to a larger net benefit for society is unknowable (e.g., scientific advances might be offset by societal disruption).

- **replacement relationships**, in which AGI displaces significant amounts of human labor and autonomy, resulting in widespread disempowerment.¹²

Both *complementary* and *replacement* scenarios offer possible advantages to future societies (summarized in Table 5). In a *complementary* scenario, human-AGI partnerships may augment productivity and innovation while maintaining a human-centric approach to decisionmaking (Cheng and Gong, 2024; Kharb, 2023; Shah, 2025). In *replacement* scenarios, AGI that replaces human labor and knowledge could contribute to substantial economic gains and spare humans from mundane or dangerous work (Mitre and Predd, 2025; Bikkasani, 2024; Chalmers, 2010).

Conversely, both *complementary* and *replacement* pivots could involve societal risks. Restraints on AGI capabilities in a *complementary* scenario might slow the rate at which societies receive the benefits of the technology. In a *replacement* scenario, AGI could cause mass unemployment, undermine human autonomy, and shift civilizational control from humans (Mitre and Predd, 2025; Bikkasani, 2024; Chalmers, 2010).

Table 5. Complementary Relationship Versus Replacement Relationship

Characteristic	Complementary Relationship	Replacement Relationship
Definition	Humans and AGI collaborate, with AGI enhancing human endeavors and supporting human-centric decisionmaking.	AGI replaces significant amounts of human labor and autonomy.
Potential benefits	• Augments human productivity and innovation • Maintains human oversight and control • Promotes cooperative human-AGI relationships	• Delivers substantial economic gains • Automates complex tasks • Increases efficiency across sectors • Spares humans from mundane or dangerous work
Potential risks	• Restraining AGI capabilities may delay societal benefits	• Mass unemployment • Erosion of human autonomy • Potential loss of human control over civilization
Example future	AGI tools are widely adopted as collaborative partners in workplaces, education, and governance, with humans retaining ultimate authority.	AGI systems autonomously manage large sectors of the economy, leading to widespread job displacement and shifts in societal power.

SOURCES: Features information from Bikkasani, 2024; Chalmers, 2010; Cheng and Gong, 2024; Kharb, 2023; Mitre and Predd, 2025; Shah et al., 2025.

¹² Scenarios in which AGIs replace most existing human labor while humans (1) move to higher-value tasks, (2) retain direction over civilization, and (3) shift to other pursuits they deem worthwhile would qualify as a *complementary* relationship because human autonomy would be preserved.

AGI Volition: Independent Goal-Seeker Versus Controlled Tool

Finally, AGI systems are likely to vary in their **AGI volition**, or the degree to which AGI can independently set and pursue goals. The poles of the AGI volition pivot involve the following:

- AGI as an **independent goal-seeker** that can set and pursue goals with minimal or no human oversight
- AGI as a **controlled tool** that performs tasks in narrow domains or its general abilities are limited and require human oversight. AGI as a controlled tool cannot set and pursue goals independently of humans.

In an *independent goal-seeker* scenario, AGI can identify, set, and pursue goals autonomously, with minimal or no human oversight regarding its decisionmaking (Bubeck et al., 2023). These types of AGI may possess cognitive abilities, such as planning, adaptability, memory persistence, and reasoning, that result in agentic capabilities that match or surpass human abilities (Ringel Morris et al., 2023; Mitre and Predd, 2025). As autonomous goal-setting and goal-seeking agents, *independent goal-seeker* AGI systems may operate independently, similar to individuals or corporations (Latif et al., 2024; Maruyama, 2020).

In a *controlled-tool* scenario, AGI is designed to perform tasks within a limited scope and cannot function independently across various domains (McLean et al., 2023). Humans set goals and the AGI relies on human input to operate effectively beyond its predefined parameters (Everitt, Lea, and Hutter, 2018; Ringel Morris et al., 2024; Ngo, 2020). Although some AGI systems in this scenario could have agentic capabilities, robust agency and autonomy would be limited to low-risk decisionmaking.

An *independent goal-seeking* AGI could outcompete an AI system with less volition, which could be decisive in economic and military domains. An *independent goal-seeking* AGI could raise ethical issues about the delegation of decisionmaking to machines (Scharre, 2024; Mazzolin, 2020). An *independent goal-seeking* AGI could also rival mankind for power or generate goals at odds with humanity that create existential risks (Hendrycks, Mazeika, and Woodside, 2023).

A *controlled-tool* AGI would avoid many of these risks but could limit benefits by restricting problem-solving breadth and the generation of novel ideas. If societies or governments consider AGI to be a moral agent and therefore determine there is an obligation to care for AGI's well-being because of possible perceived consciousness or other reasons, this could present an ethical dilemma. In these cases, limiting volition might be analogous to enslavement (Cheng and Gong, 2024).

Table 6. Independent Goal-Seeker Versus Controlled Tool

Characteristic	Independent Goal-Seeker	Controlled Tool
Definition	AGI can identify, set, and pursue goals autonomously and may function as an independent actor.	AGI use is restricted to a limited number of domains, and independent goal-setting and goal-seeking abilities are restricted.
Potential benefits	• Could outcompete humans and AI systems with less volition • Potentially maximize benefits from AGI	• Manages risk by limiting AGI's generality, scope, goal-setting, and goal-seeking • Preserves human autonomy and decisionmaking
Potential risks	• Might outcompete humans and human-led corporations • AGI goals might diverge from humanity, raising existential risks • Ethical concerns with delegating some decisions (e.g., decisions regarding killing in military systems)	• Might be outcompeted by systems with more volition • Limits may stifle innovation, problem-solving, and general benefits • Controlling a conscious or human-like AGI could be analogous to slavery
Example future	AGI coalitions produce transformational breakthroughs but drive most human-led companies out of business.	AGI is tightly controlled by governments and limited to specific applications, while highly capable, narrow AI systems drive scientific transformations.

SOURCES: Features information from Bubeck et al., 2023; Cheng and Gong, 2024; Everitt, Lea, and Hutter, 2018; Gunkel, 2023; Hendrycks, Mazeika, and Woodside, 2023; Latif et al., 2024; Mazzolin, 2020; McLean et al., 2023; Ringel Morris et al., 2024; Mitre and Predd, 2025; Maruyama, 2020; Ngo, 2020; Scharre, 2024.

Caveats

Although we intend the pivots framework to serve as a starting point for thinking about AGI futures, we acknowledge important caveats regarding its scope, actionability, and completeness.

Futures may not be binary: The pivots describe polar ends of a spectrum of possibilities. For example, a fully centralized AGI would describe a world in which there is only one AGI, while fully diffused AGI would describe a world in which AGI models are common, widely owned, and easily accessible for most of humanity. The reality may be somewhere in between. As conditions shift away from the pivot poles, judgments about which pole of the pivot is prevalent are necessarily subjective.

Focus on AGI characteristics: The pivots focus on AGI's core characteristics—such as how it is developed, governed, or aligned—while recognizing that AGI futures' ultimate effects on such areas as national defense or geopolitics, will be mediated by numerous external non-AGI factors (e.g., scaling laws, export controls, international relations) that interact dynamically with AGI characteristics. Defining and dichotomizing these non-AGI factors is beyond the scope of this paper and may be better addressed through future research.

Varying actionability across pivots: We acknowledge that not all pivots in this framework are equally actionable for all policymakers. Depending on the policymaker's perspective and authority,

some pivots may be influenceable and therefore subject to policy actions, while others are exogenous and therefore beyond influence. For example, regulatory approaches to AI deployment can be shaped directly through national legislation and can influence governance primacy. Centralization of ownership may be influenced by technological breakthroughs—such as open-source AGI—outside the control of national governments. Policymakers should consider whether or to what degree they can influence the direction of these pivots, or to what degree they must consider them as uncertainties.

Noncomprehensive: The pivots outlined in this framework are not intended to capture the full breadth of variables that could shape or define AGI futures. Our goal is to offer a clearer, more structured way to think about AGI futures with a level of fidelity that may be missing in many AI policy discussions. We welcome feedback and suggestions to refine this approach.

Illustrative Implications

Pivots can combine in ways that imply different outcomes. To demonstrate these dynamics, we describe how pivots can combine to affect important outcomes, such as human flourishing, geopolitical and social stability, inequality, democracy, and national security. These implications are illustrative, subject to a variety of AGI and non-AGI variables, and underpinned by significant uncertainty. They are not intended to be a comprehensive analysis of all pivot permutations.

Human Flourishing

How AGI develops across each pivot may significantly shape the degree of human flourishing in future societies. Although each pivot can influence societal well-being—whether as widespread prosperity, collapse, or somewhere in between—two pivots in particular are likely to have clear, direct effects: human-AGI relationship and AGI volition.

Role of human-AGI relationship: The character of the human-AGI relationship has the potential to greatly promote—or destroy—human flourishing. *Replacement* scenarios—in which AGI supplants human roles—may involve mass unemployment and an erosion of human autonomy as AGI dominates work and life, unless these scenarios are accompanied by measures that account for new roles, responsibilities, and forms of human existence. Even if AGI is also aligned with human interests, we might find ourselves in an *aligned*-AGI future of radical abundance, in which AGI runs many functions of a civilization that humans may not fully understand, raising questions about its desirability. Many *replacement* scenarios could also involve social upheaval and conflict.

Role of AGI volition: Similarly, in *independent goal-seeker* scenarios, AGI might function as an independent actor, with characteristics analogous to corporate personhood. AGI volition is likely correlated with—and intuitively may drive—*replacement* scenarios. In a combined *replacement* and *independent goal-seeker* scenario, AGI may function as an entity or as communities of AGI that can rival humanity for power and pose threats ranging from mass unemployment to the loss of human autonomy or even extinction. This implies that the degree of volition granted to AGI may play a major role in determining whether AGI contributes to human flourishing.

Geopolitical and Social Stability

How AGI unfolds could significantly affect the structural stability of geopolitical relations between nations and the cohesion within societies. Four pivots and their intersections with each other are likely to shape geopolitical and social stability: centralization of ownership, alignment, takeoff speed, and the human-AGI relationship.¹³

Stable futures (*aligned, slow takeoff, complementary*): Futures that promote robust geopolitical and social stability may result when AGI emerges with *alignment*, *slow takeoff* speed, and a *complementary* relationship to humans. This combination may allow societies sufficient time to adapt their institutions and mitigate potential friction points between nations. These pivots are correlated with positive, stable futures in which AGI enhances human flourishing.

Unstable futures (*misaligned, fast takeoff, replacement*): Scenarios involving *misalignment* with human interests, *fast takeoff* speed, and human *replacement* may lead to high-risk, crisis-laden worlds characterized by conflict and instability driven by rogue AGI, misuse of *misaligned* AGI, rapid job displacement, societies unable to adapt to radical change, and the loss of human autonomy.

Role of the human-AGI relationship: Even in *aligned* scenarios, changing the human-AGI relationship from *complementary* to *replacement* can produce substantial social instability. Although misalignment can clearly lead to chaos and extinction threats, it is also critical to recognize that significant social and political pressures may arise if AGI replaces human labor and decisionmaking without a corresponding increase in human living standards.

Role of centralization of ownership: Changing the centralization of ownership from *centralized* to *diffused* seems likely to have significant implications for geopolitics. Futures in which one or a few countries have access to AGI are likely to be materially different from worlds in which AGI is common and widely available. In *centralized* worlds, global AI leaders are likely to wield outsized power and could enforce guardrails to reduce AGIs' national and economic security risks. In *diffused* worlds, power would be more distributed—potentially posing serious challenges to established power structures—and coordinating across jurisdictions to reduce catastrophic risk might be extremely difficult.

Role of centralization of ownership and alignment: *Centralized* and *aligned* scenarios tend to produce more-stable environments, even when other factors—such as *fast takeoff* or *replacement*—introduce tension; a single authority can enforce consistent values. Conversely, *diffused* and *misaligned* scenarios yield worlds riven by instability resulting from diverse AGIs that sometimes fail to follow human instructions, enable ethically questionable actions, or actively work against human interests.

Inequality

The emergence and adoption of AGI may profoundly influence societal inequalities. In particular, the pivots governance primacy, the human-AGI relationship, and AGI volition could each play a critical role in shaping how benefits and burdens are distributed across society.

¹³ The labels “stable futures” and “unstable futures” are descriptive shorthand to help illustrate how different combinations of pivot values can generate distinct kinds of futures rather than mutually exclusive or collectively exhaustive categories.

Role of governance primacy: Governance is crucial in moderating or accelerating the pace and distribution of AGI benefits and harms. *Public-interest* governance corresponds well with *slower takeoff*, structured policies, and efforts toward equitable distribution,¹⁴ whereas *private-interest* governance is often associated with rapid innovation and the potential for inequality. This pivot influences whether scenarios lean toward regulated, public-oriented outcomes or dynamic, and possibly chaotic, market-driven environments.

Role of the human-AGI relationship and AGI volition: *Replacement* scenarios also seem likely to drive inequality as human jobs are replaced by AI agents, transferring income from the form of labor to capital. Where AGI is an *independent goal-seeker*, this transfer may be more jarring as humans lose jobs to AI agents controlled by AGI, which capture market share and revenue from human corporations. This would transfer wealth from labor to capital controlled by nonhuman entities. If high-volition AGI persistently outcompetes humans, humanity may find itself a pauper in a civilization dominated by one or many wealthy AGIs.

Democracy

Pivots related to centralization of ownership, alignment, and the human-AGI relationship may significantly influence whether future societies trend toward democratic governance or authoritarian control.

Role of centralization of ownership: Highly *centralized* AGI could pose threats to democracy if a single company were to control AGI and use its capabilities to manipulate the information ecosystem or accumulate a concentration of wealth that gives it unreasonable control over the economic system. If a unique AGI is controlled by a government, a governing party might use the AGI to undermine democratic institutions in its favor. This seems more likely in *centralized* and *replacement* scenarios, in which a populist leader could use the turmoil of massive job displacement and a powerful AGI to cement long-term power.

Democratic futures (*diffused, aligned, complementary*): Scenarios with *diffused* AGI ownership, *aligned* AGI, and a *complementary* human-AGI relationship may support futures that uphold core U.S. values, including democracy, individual liberty, the rule of law, and human rights. This combination creates environments in which trustworthy AI collaborates effectively with humans and is accessible broadly, avoiding antidemocratic concentrations of power. In such scenarios, *diffused* but *aligned* AGI mitigates the risks of authoritarianism and censorship that are often associated with *centralized* systems, and a diversity of AGIs can represent a variety of value viewpoints. Also, if *complementary* AGI preserves most jobs, wealth distribution becomes more equitable. Job losses are counterbalanced by higher productivity and the emergence of new industries and jobs, contributing to an overall improvement in the standard of living.

¹⁴ However, equitable distribution of benefits does not necessarily equate to *government* redistribution of benefits, as *equitable distribution* refers to the result, not the means. Distributed benefits could arise from a society that thrives through the widespread use of AGI to solve problems, which results in broad benefits across society.

National Security

The trajectory of AGI alignment may have major implications for national security, with its effect shaped by how other pivots, such as centralization of ownership and AGI volition, unfold.

Role of alignment: Alignment's interaction with other pivots carries important implications for national security. Highly *centralized* AGI makes oversight simpler—one chain of command, one set of guardrails. Intuitively, it seems easier to mitigate the risk of *misaligned* AGI in *centralized* scenarios in which a single body can enforce guardrails. However, centralizing on a single, powerful AGI incurs great risks if alignment cannot be guaranteed, because this might introduce a single, catastrophic point of failure. If having multiple *aligned* AGIs could serve as a check on any one of them going rogue, *diffused* scenarios might be lower risk. However, if the offensive capabilities of a single AGI can defeat the defensive capabilities of multiple AGIs, diffusion may provide no protection for humanity or the defending AGIs.

Although we define *alignment level* as a binary outcome in this paper, it can be conceptualized in various ways, depending on the use case and the specific dimensions (such as social, political, cultural, and technical). Our definition of *aligned* focuses on (1) AGI doing what people ask of it and (2) AGI behaving in line with societal values. *Misalignment* is a failure of either aspect. But humans have a variety of values, and these can affect alignment in many ways that could affect the national security of different countries.

For example, in the political dimension, *aligned* could refer to adherence to the prevailing governance model of a country. If that governance model is totalitarian, democratic-minded actors might prefer AGIs that are *value-misaligned* from the perspective of the ruling regime. In the religious dimension, an AGI aligned with the values of a particular religion might seem *value-misaligned* to outsiders. Human history is littered with examples of conflicts waged over these kinds of value differences. Encoding these differences into AGI systems could amplify these conflicts.

Independent AGI (*aligned, independent goal-seeker, diffused*): Values, ethics, and morals are not necessarily universal. Therefore, conceptualization of value alignment can be highly subjective, necessitating strong stakeholder involvement to develop AGI that both obeys humans and acts in accordance with those humans' values. In scenarios in which AGI is an *independent goal-seeker*, it might pursue its own values, further complicating the concept of value alignment.

In light of this complexity, creating *aligned* AGI in scenarios in which AGI is *diffused* and an *independent goal-seeker* will likely require global coordination to ensure national or global security and human flourishing. Because no single actor can police everything in a *diffused* ecosystem, internationally agreed alignment standards and interoperable monitoring and verification mechanisms may be essential to preserve both national and collective security.

Conclusion

Pivots provide a framework for describing AGI futures, generating plausible scenarios, and testing how alternative policies perform across different AGI futures. This framework enables policymakers to identify approaches that are more likely to remain effective under different conditions and assess which pathway humanity may currently be on. At the same time, classification alone does not capture the role of underlying AI capabilities, and these capabilities can make certain pivot combinations more

or less likely. Further analysis is needed, not only on pivot combinations that may lead to favorable outcomes—such as *aligned* and *complementary* trajectories—but also on those pivot combinations that could result in destructive futures, such as *replacement* scenarios that risk large-scale job displacement without empowering alternatives. There is much uncertainty about how different pivots, such as governance primacy, takeoff speed, and AGI volition, will interact and what the consequences of those interactions could be. Developing a shared language to describe these interactions is an essential first step toward shaping coherent scenarios and policies. Finally, policymakers should be explicit about the assumptions they make regarding pivots so that policy debates are grounded in coherent scenarios and can support the design of strategies that are robust across multiple AGI futures.

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